EXAM – CHALLENGE 2: Investigating Intraocular Pressure (IOP) Variability in Primary Open-Angle Glaucoma (POAG)

Primary open-angle glaucoma (POAG) is one of the leading causes of blindness in the United States and worldwide. Three to six million people in the United States are at increased risk for developing POAG because of elevated intraocular pressure (IOP). Imagine you are a researcher contributing to a study investigating how variability in intraocular pressure (IOP) affects the development and progression of primary open-angle glaucoma (POAG). Your specific task is to develop a mini-research project focusing on the long-term variability of IOP and its potential impact on POAG.

Part 1: Introduction
Write a brief overview of your research proposal, addressing the following:

- Research Question: Define a clear, focused research question that your study will address.
- Hypothesis: Based on prior findings, state your hypothesis. For example, how do you expect IOP variability to affect the onset of POAG?
- Rationale: Explain why this research question is important. Refer to existing controversies in the literature regarding IOP variability and its independent role in POAG progression beyond baseline IOP.

Part 2: Materials and Methods
Develop a basic outline for the research methods you would use, considering the following components:

- Study Population and Sample: Describe the characteristics of your study participants (e.g., number of participants, inclusion criteria, and key demographics such as age or ethnicity).
- Covariates: List relevant covariates you would control for in your analysis (e.g., baseline IOP, age, ethnicity, ...). Describe why these are important for interpreting results accurately.
- Outcome Measurement: Describe the main outcome you would examine.

Describe the statistical approach you would use to analyze your data:

- Statistical Models: Specify the types of regression models (e.g., Cox proportional hazards) you would employ to analyze the relationships between IOP variability and glaucoma outcomes.

Part 3: Results
In the Results section of your report, summarize the key findings based on the dataset provided.
Describe any associations you found between IOP variability and the outcome. Highlight if higher IOP variability is generally linked with faster increased risk of glaucoma, or if the variability impact differs across patient subgroups.

Summarize the potential implications of these results, particularly regarding how IOP variability might influence progression risk for patients across different individuals. Your results should be presented clearly and concisely, focusing on observed patterns in the data without making conclusions or policy recommendations, which will be addressed in the Discussion section.

Part 4: Discussion and Conclusions
Write a brief discussion of your research findings and how they might contribute to public health, clinical practice, or glaucoma management strategies. Consider how your findings could inform future treatment guidelines or monitoring protocols for POAG patients.

Grading Criteria:

1. Clarity and focus of research question and hypothesis
2. Coherence between python code and the methodology described in the paper
3. Relevance and depth of methodology
4. Appropriateness and rigor of statistical analysis approach
5. Insightfulness of discussion and policy recommendations

Dataset Variable Descriptions:

A cohort of 819 patients from USA has been followed-up for 10 years. These patients were at risk of developing POAG, since they had previously at least one measurement of IOP >= 24 mmHg, but they were not yet diagnosed with POAG and they were not treated for POAG during the study or before.

A series of variables were measured at the beginning of the study (baseline), while periodic follow-up examinations measured IOP, visual field and optic disc deterioration. These periodic IOP measures determined the median, std, range and max values of IOP during follow-up.

The primary outcome was the development of reproducible visual field abnormality or reproducible optic disc deterioration attributed to POAG. Abnormalities were determined by masked certified readers at the reading centers, and attribution to POAG was decided by the masked Endpoint Committee. These outcome is represented with status=1 in the dataset, while status=0 indicates censoring.

The variables included in the dataset are the following ones:

1. Age (baseline)
2. Sex (1=Male 0=female) (baseline)
3. Ethnicity (White, Black, Hispanic, Asian, Native American, Other) (baseline)
4. Hypertension (1 or 0) (baseline)
5. Diabetes (1 or 0) (baseline)
6. Migraine (1 or 0) (baseline)
7. CVD (Cardiovascular Disease) (1 or 0) (baseline)
8. Familiarity (for glaucoma) (1 or 0) (baseline)
9. corneal_thickness (in micrometers) (baseline)
10. vcdr (vertical cup-to-disc ratio) (baseline)
11. Base_IOP (baseline)
12. FUP_IOP (mean IOP during follow-up)
13. SD_IOP (standard deviation of IOP during follow-up)
14. Range_IOP (range of IOP during follow-up)
15. Max_IOP (max IOP during follow-up)
16. CV_IOP (coefficient of variation for IOP = SD_IOP / FUP_IOP)

All IOPs are measured in mmHG
Note that familiarity with glaucoma is a known risk factor for the disease, while for the listed comorbidities, in particular hypertension, there are contrasting opinions in the literature. Corneal thickness and VCDR values might have links with glaucoma, but they are not primarily used to diagnose glaucoma.

Deadline: 18 December 2024 11:00 CET